BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES

COURSE HANDOUT

Part A: Content Design

Course Title	Information Retrieval
Course No(s)	AIML/DSE CZG537
Credit Units	4
Credit model	3-1-0
Course Author	Dr. Chetana Gavankar and Dr.Vijayalakshmi
Version No	2.0
Date	20-02-2025

Course Description:
Organization, representation, and access to information; categorization, indexing, and content analysis; data structures for unstructured data; design and maintenance of such data structures, indexing and indexes, retrieval and classification schemes; use of codes, formats, and standards; analysis, construction and evaluation of search and navigation techniques; search engines and how they relate to the above. Multimedia data and their representation and search.

Course Objectives

No	Course Objective
CO1	To understand the structure and organization of various components of an IR system
CO2	To understand information representation models, term scoring mechanisms, etc. in the complete search system
CO3	To understand the architecture of search engines, crawlers, and the web search
CO4	To understand cross-lingual retrieval and multimedia information retrieval

Text Book(s)

T1	C. D. Manning, P. Raghavan and H. Schutze. Introduction to Information Retrieval, Cambridge University Press, 2008. http://nlp.stanford.edu/IR-book/

Reference Book(s) & other resources

R1	Modern Information Retrieval, Ricardo Baeza-Yates and Berthier Ribeiro-Neto, Addison-Wesley, 2000. http://people.ischool.berkeley.edu/~hearst/irbook/
R2	Ricci, F.; Rokach, L.; Shapira, B.; Kantor, P.B. (Eds.), Recommender Systems Handbook. 1st Edition., 2011, 845 p. 20 illus., Hardcover, ISBN: 978-0-387-85819-7

R3	Cross-Language Information Retrieval by Jian-Yun Nie Morgan & Claypool Publisher series 2010
R4	Multimedia Information Retrieval by Stefan M. Rüger Morgan & Claypool Publisher series 2010.
R5	Web Data Mining: Exploring Hyperlinks, Contents, and Usage Data by B. Liu, Springer, Second Edition, 2011.
R6	Abhinav Kimothi (2024). *A Simple Guide to Retrieval Augmented Generation.* Manning Publications.
R7	Hang Li (2014). *Learning to Rank for Information Retrieval and Natural Language Processing.* Morgan & Claypool Publishers.
R8	Olivier Sigaud, David Filliat, and Emmanuel Dellandréa (2023). *Multimodal Machine Learning: Techniques and Applications.* Springer.

Learning Outcomes:

No	Learning Outcomes
LO1	Students will gain an understanding of an information retrieval system as a whole and about its components.
LO2	Students will have knowledge about the design issues and their solutions of different types of models including Boolean, vector space, etc.
LO3	Students will have a detailed understanding of text indexing, mining, weighting schemes, etc.
LO4	Students will acquire knowledge about cross-lingual and multimedia information retrieval.
LO5	With the acquired knowledge students will be able to design and build different kinds of information retrieval systems.

Modular Content Structure

1. Introduction
    1.1. Information Retrieval
    1.2. Basic Search Model
2. Basic Information Retrieval Concepts
    2.1. Boolean Retrieval
    2.2. Dictionaries and Tolerant Retrieval
    2.3. Index Construction and Compression
3. Vector Space Model
    3.1. Scoring, Term Weighting
    3.2. The Vector Space Model for Scoring
4. Text Mining
    4.1. Text Classification

4.2. Vector Space Classification

4.3. Text Clustering_K-Means Clustering

5. Evaluation in Information Retrieval

5.1. Evaluation in unranked retrieval sets

5.2. Evaluation in ranked retrieval sets

6. Web Search

6.1. Web Search Basics

6.2. Web Crawlers and Indexes

6.3. Link Analysis

6.4. Learning to Rank Models

7. Cross-Lingual Retrieval

7.1. Language Problems in IR

7.2. Approaches for CLIR

7.3. Neural

8. Multimedia Information Retrieval

8.1. Multimedia Search Technologies

8.2. Content-Based Retrieval

8.3. Multimodal Retrieval and Vision–Language Models (CLIP, ALIGN)

9. Recommender Systems

9.1. Collaborative and Content-Based RS

10 Neural IR

10.1 Introduction to Deep neural network

10.2 Deep neural network for IR

10.3 **Generative IR and RAG**

10.4 **Conversational and QA-based IR (ChatGPT, GPT-based Search)**

Part B: Contact Session Plan

Academic Term	Mtech DSE June 2021
Course Title	Information Retrieval
Course No	AIML/DSE CZG537
Lead Instructor	Dr. Chetana Gavankar

Contact Session	List of Topic Title (from content structure in Part A)	Topic # (from content structure in Part A)	Text/Ref Book/external resource

1	• Introduction o Information Vs Data Retrieval o Basic Concepts o The retrieval process o Taxonomy of IR o Classic IR and Alternative models	1.1, 1.2	R1 Ch1, Ch2
2	• Boolean Retrieval o Inverted index o Processing Boolean queries o Boolean Vs Ranked retrieval o Term vocabulary and postings lists o Phrase queries	2.1	T1 Ch 1, 2
3	• Dictionary and Tolerant Retrieval o Search Structures for dictionaries o Wildcard queries o Phonetic Correction	2.2	T1 Ch3
4	• Index Construction and Compression o Blocked sort-based Indexing o Single pass in-memory indexing o Distributed and dynamic indexing o Dictionary comparison o Postings file compression	2.3	T1 Ch4,5
5	• Vector Space Model o Term frequency and weighting o The vector space model for scoring o Tf-idf functions	3.1, 3.2	T1 Ch6
6	• Classification & Clustering for IR o Feature Selection o Vector space classification o Document Representation o Rocchio classification o Evaluating Classification o Text Clustering_K-Means Clustering	4.1	T1 Ch13, 14 T1 Ch16, 17
7	• Evaluation in Information Retrieval o Evaluation in unranked retrieval sets o Evaluation in ranked retrieval sets		T1 Ch 8
8	• Review		
9	• Web Search o Web characteristics o The search user experience o Index size and estimation o Learning to Rank models • Types of learning to rank	5.1	T1 Ch19 R7 and document will be shared by Faculty

	models		
10	• Web Crawling and Indexes o Crawling o Crawler Architecture o Distributed Indexes	5.2	T1 Ch20
11	• Link Analysis o The web as a graph o Google's page rank o Hub and Authorities (HITS)	5.3	T1 Ch21
12	• Cross Lingual IR (CLIR) o Language problems in IR o Translation Approaches o Handling Many Languages o Resources for CLIR	6.1, 6.2	R3 Ch2
13	• Multimedia IR o Basic Multimedia search technologies o Content Based Retrieval o Multimodal Retrieval and Vision–Language Models (CLIP, ALIGN) o Architecture of CLIP and ALIGN o Contrastive learning objective o Zero-shot learning capability o CLIP embedding for retrieval	7.1,7.2	R4 Ch2,3 R8 and documents will be shared by Faculties for CLIP and ALIGN
14	• Recommender System o Collaborative recommendation o Content based recommendation o Other type & hybrid recommendations	8.1	R2 Ch1-5
15	NEURAL IR Introduction to Deep neural networks o Input text representation o Standard architectures Deep neural networks for IR o Document autoencoders o Siamese networks o Interaction-based network o Lexical and semantic matching o Matching with multiple document fields o Generative IR and RAG o Conversational and QA-based IR (ChatGPT, GPT-based Search)	6&7	An Introduction to Neural Information Retrieval by Bhaskar Mitra, Nick Craswell Link: https://www.microsoft.com/en-us/research/uploads/prod/2017/06/fntir2018-

			neuralir-mitra.pdf
16	• Review		

Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc...) number of lab exercises needed, etc.

1. Lab work: 6
2. Project work: 0
3. Case Study: 4 Webinars
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

To provide hands-on experience in implementing fundamental Information Retrieval algorithms using state-of-the-art tools and frameworks, enabling students to bridge theoretical concepts with practical applications in modern retrieval systems.

Scope of Experiential Learning Component:

Programming language - Python

Tools and libraries: NLTK, Numpy, Beautiful Soup, PyTorch/Tensor flow

Lab Infrastructure:

Online/ Open source/Google Colab

Detailed Plan for Lab work

Lab	Lab objective		Session Reference
1	Text Preprocessing		1
2	Boolean Retrieval model		2
3	Constructing an Inverted Index		4
4	Vector space model		5

5	Web crawler to retrieve news stories		10
6	Page ranking		11
7	Cross lingual Information Retrieval		12
8	Recommender system using Collaborative Filtering		14

Evaluation Scheme:
Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

No	Name	Type	Duration	Weight	Day, Date, Session, Time
EC-1(a)	Quizzes	Online		10%	TBA
EC-1(b)	Assignments	Take Home		20%	TBA
EC-2	Mid-Semester Test	Closed Book		30%	TBA
EC-3	Comprehensive Exam	Open Book		40%	TBA

Note:
Syllabus for Mid-Semester Test (Closed Book): Topics in Session Nos. 1 to 8
Syllabus for Comprehensive Exam (Open Book): All topics (Session Nos. 1 to 16)

Important links and information:

Elearn portal: https://elearn.bits-pilani.ac.in or Canvas
Students are expected to visit the Elearn portal on a regular basis and stay up to date with the latest announcements and deadlines.

Contact sessions: Students should attend the online lectures as per the schedule provided on the Elearn portal.

Evaluation Guidelines:

1. EC-1 consists of two Quizzes. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
2. EC-2 consists of either one or two Assignments. Students will attempt them through the course pages on the Elearn portal. Announcements will be made on the portal, in a timely manner.
3. For Closed Book tests: No books or reference material of any kind will be permitted.
4. For Open Book exams: Use of books and any printed / written reference material (filed or bound) is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
5. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam which will be made available on the Elearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course hand-out, attend the online lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the hand-out.